

Understanding Human Perception of Bus Fullness: An Empirical Study of Crowdsourced Fullness Ratings and Automatic Passenger Count Data

Xidong Pi¹, Zhen (Sean) Qian^{1,2}, Aaron Steinfeld³, and Yun Huang⁴

Transportation Research Record
1–10

© National Academy of Sciences:
Transportation Research Board 2018

Reprints and permissions:

sagepub.com/journalsPermissions.nav

DOI: 10.1177/0361198118781398

journals.sagepub.com/home/trr



Abstract

Bus fullness, namely bus crowding level, is one of the most critical metrics used to quantify public transportation service quality and consumer satisfaction. Often it is simply represented by the head count within a bus vehicle that is measured using the prevailing automatic passenger counter (APC). Little is known, however, about how the precise passenger count reflects the human perception of bus fullness. This study examines the linkage between APC data and crowdsourced fullness ratings data from riders in Pittsburgh, U.S.A., to understand how riders' perception of bus fullness is related to spatial, temporal, and demographic factors in addition to the passenger counts. We found that human perception of bus fullness, matched with passenger counts, can vary substantially by time of day, bus vehicle seat capacity, and passengers' income. Last but not least, we proposed a statistical model that estimates riders' perception of bus fullness based on APC data and bus route characteristics. This model can be used by transit agencies who possess APC data to assess service quality better, and ultimately to enhance the design and operation of transit systems.

Bus fullness, namely bus crowding level, is one of the most critical metrics used to quantify the quality of and customer satisfaction with public transportation services (1, 2). Bus crowding results in not only uncomfortable travel experience but also low efficiency of bus service. It increases dwell time at bus stops (3), and overcrowding and discomfort on buses can double the average cost of trips (4). Many public transit agencies use the loading level of a bus vehicle to represent the bus fullness, defined by the ratio of current loads (passenger counts) over the vehicle seat capacity. It is common practice to use automatic passenger counter (APC) systems to compute this ratio as the metric of bus fullness. The prevailing APC systems (5, 6), usually accompanied by automatic vehicle location (AVL) systems, help automate the measurements of passenger counts in every trip and at every stop, which was traditionally collected by hired observers at a much smaller coverage of spatial and temporal scale. By deploying APC systems, public transit operators and planners are now receiving and archiving considerably more passenger count measurements than in the past.

However, this definition of bus fullness does not take into account riders' perceptions, which may depend on various factors. First, rider perception of bus fullness varies by individual. Some users may regard the same level of fullness as more crowded than others. Thus, they may feel more discomfort than others on the same vehicle. The perception of bus fullness

may also vary by time of day and location. Users may psychologically feel more uncomfortable at certain times and locations, and thus perceive the effect of fullness very differently. For instance, peak hours may often cause a higher crowding level than off-peak hours, but passengers are also more used to overcrowding in peak hours. Given the same ridership over seat capacity ratio, a peak-hour bus may be perceived as less crowded than an off-peak bus.

To understand bus fullness better in order to quantify service quality and customer satisfaction, we examine human perception of bus fullness instead of simply using the ratio of passenger count over seat capacity. In particular, we answer the following questions: How much is there a gap between machine measurements (APC data) and human perception of bus fullness? Is there any discrepancy in the perception among different bus riders? What factors are associated with human perception of fullness? Can we infer human perception from APC measurements combined with other factors?

¹Department of Civil and Environmental Engineering, Carnegie Mellon University, Pittsburgh, PA

²Heinz College, Carnegie Mellon University, Pittsburgh, PA

³Robotics Institute, Carnegie Mellon University, Pittsburgh, PA

⁴School of Information Studies, Syracuse University, Syracuse, NY

Corresponding Author:

Address correspondence to Zhen (Sean) Qian: seanqian@cmu.edu

Thanks to the Tiramisu project (7–10), a crowdsourced transit platform providing real-time bus location and fullness in the Pittsburgh region, we have access to bus fullness ratings from app users that reflect their perceptions. Users of the Tiramisu application submit their perceived ratings of bus fullness at their boarding bus stops. The ratings are categorized by “many seats”, “few seats”, “stand,” and “full”. This paper examines both the crowdsourced fullness ratings data from Tiramisu and APC data to understand how riders’ perceptions of bus fullness are related to spatial, temporal, and demographic factors in addition to the measured passenger counts. Few transit agencies outside of Pittsburgh are able to measure fullness perception directly from riders. Therefore, they can use APC data to infer the perceived fullness to assess the service quality better, and ultimately to enhance system design and operation.

This research contributes to the literature in the following aspects:

- Provides an empirical study of the relation and comparison between APC data and crowdsourced ratings of bus fullness (a proxy of human perception of bus fullness).
- Shows the effect of several factors on human fullness perception. Those factors include time of day (e.g., peak hours versus off-peak hours), bus seat capacity (e.g., 40-seats versus 60-seats), household income level.
- Develops an estimation model to infer passengers’ perception of fullness from APC measurements combined with other spatial, temporal, and demographic factors.

The paper is organized as follows. In the next section, we first provide a literature review on studies of bus load and bus fullness, followed by an introduction to the characteristics of our two data sets: APC data and crowdsourced fullness ratings, as well as the data preprocessing methods. Then, we present the statistical experiment results and analyses. We also provide a brief discussion of our findings. The final section presents the conclusions from this research.

Literature Review

Research on bus fullness is fairly intensive. Concerning passengers’ perceptions, teams have studied how bus crowding is related to passenger satisfaction using case studies and survey data (2, 11). From a quantitative perspective, other research has investigated how bus crowding impacts the transit service quality in terms of the dwelling time and service reliability (12, 13), and predicted the bus fullness and passenger demand (14, 15). The research gap is that the measured bus fullness (or ridership to capacity ratio) can differ from passengers’ perceptions of fullness, and the relationship is still unclear.

Li and Hensher (16) define two types of measures of crowding: measured passenger ridership as an “objective” measure and passenger’s perceived crowding as a “subjective” measure. These two measures are not identical, and the perceived crowding is a compound of actual passenger density and many other factors, such as personal preferences

and environmental characteristics. The authors also provide a comprehensive discussion about the discrepancy between measured and perceived crowding. Others have shown that this discrepancy is significant through two recent surveys from a rail operator in Australia (17). Rail passengers’ perceptions of crowding are also related to psychological factors. The authors proposed a self-rating questionnaire to study the users’ perceptions of crowding on rail and found a complex connection among passenger stress, negative feelings, and crowding. Nevertheless, most transit agencies and public transit literature still use the “objective” bus fullness measures, directly from APC data (18, 19). Therefore, it is critical to study the relations between “objective” and “subjective” bus fullness quantitatively.

In this paper, we clearly identify two measures of bus fullness: APC-based passenger ridership to capacity and crowdsourced user ratings on fullness. The crowdsourced user fullness ratings from mobile applications enable us to study the subjective and objective bus fullness measures statistically. We also build an estimation model (a coarse mapping between the two dimensions) based on a large volume of data.

Data Description

Crowdsourced Fullness Data

Tiramisu (7–9) is a mobile phone application which was designed to crowdsource and share bus information, such as real-time bus location, arrival time, bus fullness, issues, and riding stories in the community. Users of the Tiramisu application share their GPS location and submit the fullness ratings in four categories: “many (seats),” “few (seats),” “stand,” or “full”, when they are boarding. This application is able to provide the estimated vehicle arrival time and user-perceived bus fullness information in real-time to other users. It helps to improve the accessibility of the transit service and the user’s experience. Tiramisu was the only way for passengers to get real-time transit information in Pittsburgh before the AVL/APC systems were deployed at the end of year 2012.

In this research, we use the crowdsourced fullness ratings data of all weekdays from September 2012 to October 2015 from Tiramisu app (except for August 2013, October 2013–January 2014, and November 2014, because either the APC system or Tiramisu app was off-line during those time periods). Each rating represents the user’s perception of bus fullness at the boarding stop. Each rating also comes with a trace in the data set that contains information regarding trip date, trip ID, route ID, boarding location, bus stop, and time-stamp, as well as the user’s fullness rating. These fullness ratings covered all major bus routes operated by the Port Authority of Allegheny County (PAAC), the sole transit provider of Allegheny County.

APC Data

The APC data used in this research were collected and provided by PAAC. PAAC upgraded and installed the APC system on all buses in 2012. It measures the number of passengers

boarding and disembarking by means of laser sensors at each bus door, and then archives the data in on-board computers. The system is activated immediately when a bus departs its depot and starts a trip. The APC data contains trip information (date, route ID, direction [inbound or outbound], trip ID, the number of seats), stop information (stop ID, arrival and departure time), and passenger counts (boarding, alighting, and in-vehicle counts). We compute the bus fullness for any bus trip at any location and time following:

$$\text{Fullness} = \frac{\text{Number of Passengers}}{\text{Number of seats}}$$

We refer to this metric as “APC-based fullness” or “passenger-over-capacity ratio” throughout this paper. In Pittsburgh, it is a real number spanning (0.0, 2.0) in most cases, with the very rare occurrence of being greater than 2.0. This implies that almost all bus loads in Pittsburgh were less than 200% of seat capacity.

Data Preprocessing

The crowdsourced data were first pre-processed to remove the records with empty fullness ratings. We then filtered out users with less than or equal to three contributions to eliminate fraudulent and inaccurate submissions from inactive users. By matching the date, route ID, direction, trip ID, and stop ID, we joined both APC and fullness ratings data, which essentially

created one-to-one mapping from fullness ratings to APC-based fullness for bus trips and stops. After this mapping, we consolidated the two crowdsourced fullness rating classes “stand” and “full” to one class “full”, due to the possible linguistic ambiguity between “stand” (a bus has no available seats) and “a bus is full (full capacity)”. In addition, the number of records for these two classes was much less than for the other two classes “many seats” and “few seats.” Consolidating the two classes may enhance statistical significance. Finally, we filtered out “irresponsible” users, which were defined as users with more than 15% of “outlier” ratings. An outlier rating was a rating with its associated APC-based fullness value smaller than 5 quantiles or greater than 95 quantiles of all samples at this rating. For example, if the mean APC-based fullness of all samples rated as “many seats” is 0.30, and the 5 and 95 quantiles were 0.03 and 0.80 respectively, then samples that were rated as “many seats” records and their associated APC-based fullness value less than 0.03 or greater than 0.80 were identified as outlier ratings.

After data preprocessing, there were in all 42,238 valid rating traces from 1,529 users, plotted in Figure 1. It showed that the rating data covered all main corridors and streets along bus routes. For each of the three fullness rating classes:

- Many seats: 23,092 valid records
- Few seats: 13,569 valid records
- Full: 5,577 valid records

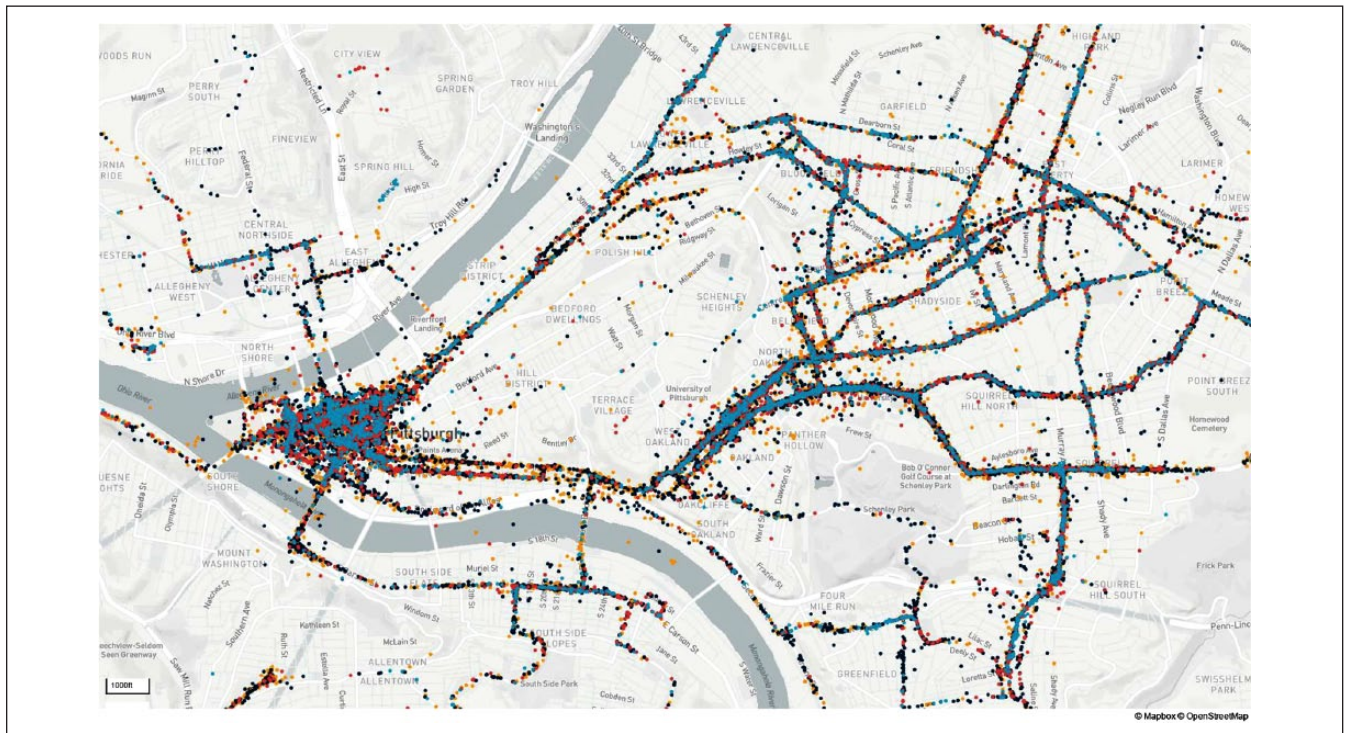


Figure 1. Crowdsourced bus fullness ratings data: spatial distribution in central districts of Pittsburgh. Year 2012= orange; 2013= black; 2014= pink; 2015= cyan. Created with ©Mapbox (20).

Experiments and Results

Statistical Difference between Crowdsourced Fullness Ratings and APC-Based Bus Fullness

The first question we wanted to answer was whether user fullness ratings were statistically consistent with the actual bus load. We created a hexbin plot of time of day (x-axis) and APC-based measurements (y-axis) in Figure 2(a–c) for each of the three rating classes: “many,” “few,” and “full,” along with marginal distributions. The boxplot

of APC measurements for the three rating classes is in Figure 2(d). On average, users perceived around 31% of bus loads over capacity to be “many seats,” around 52% to be “few seats,” and around 76% to be “full.” However, the variance increases when the fullness is rated as high. This indicates human perception is generally more consistent when a bus is empty, whereas the perception is more diversified when the bus is almost full. Also notice that the marginal distributions of APC-based fullness by time of day are more concentrated towards morning and

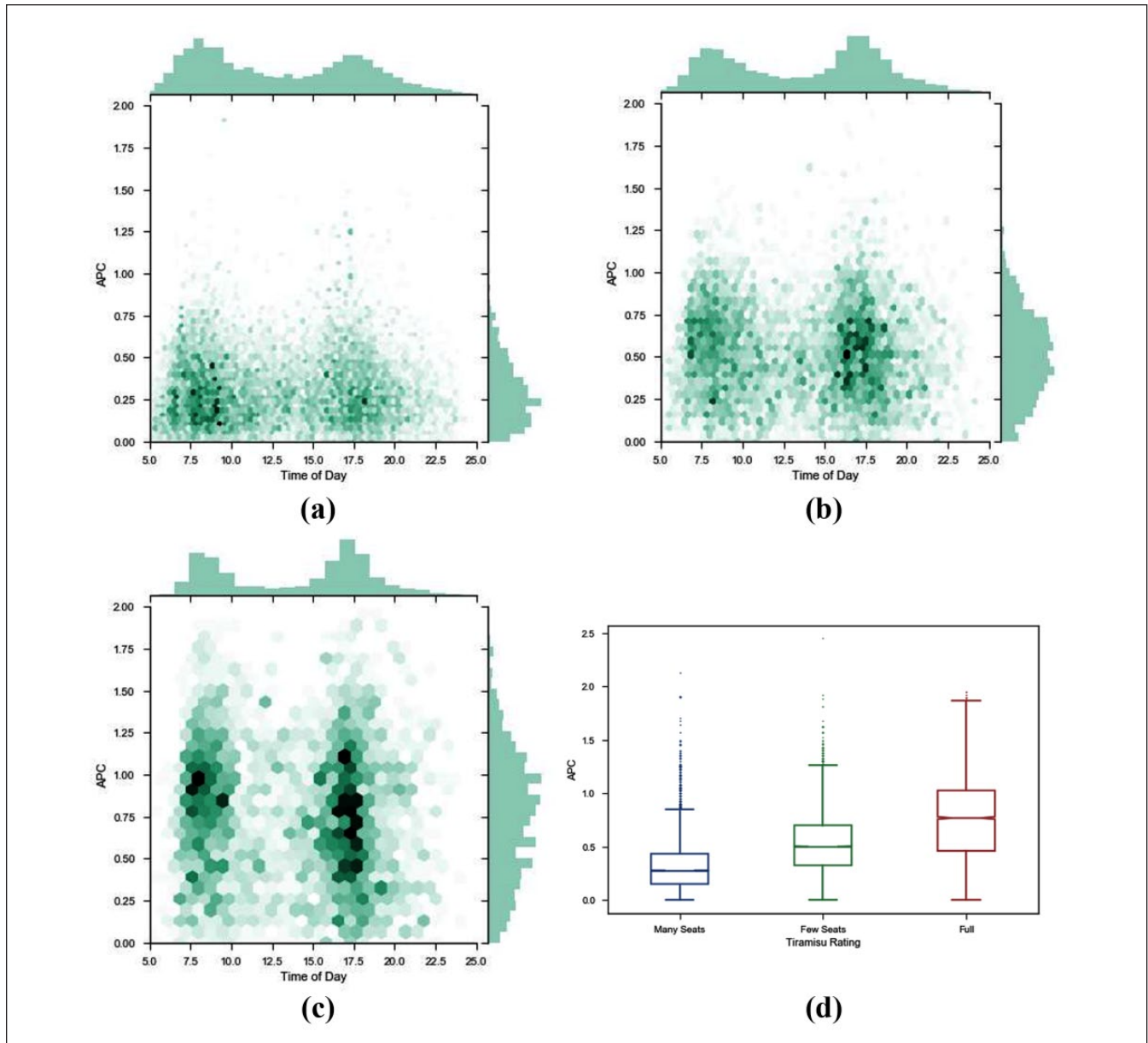


Figure 2. The hexbin plot and box plot of time of day and APC-based fullness. (a) “many seats”, mean = 0.315, SD = 0.218; (b) “few seats”, mean = 0.517, SD = 0.270; (c) “full”, mean = 0.758, SD = 0.382; (d) box plot of APC fullness at each rating level.

Table 1. Statistical Tests for the Three Rating Groups

Groups	One-way ANOVA test		Kruskal–Wallis <i>H</i> -test	
	Statistics	<i>p</i> -value	Statistics	<i>p</i> -value
All 3 groups	5014.4	<0.005*	7114.0	<0.005*
“Few” versus “many”	4213.2	<0.005*	3946.9	<0.005*
“Many” versus “full”	9328.1	<0.005*	5172.4	<0.005*
“Few” versus “full”	2437.7	<0.005*	1719.3	<0.005*

afternoon peak hours at the higher rated fullness, implying more consistent human perception/ratings for a full bus than an empty bus during peak hours. However, the box plot also implies that aggregating over all time of day, human perception is generally more consistent among all users when the bus is empty than it is full.

We performed a one-way ANOVA on the three groups of data: “many seats”, “few seats,” and “full”, to test if the means of the APC-based fullness among the three groups are different statistically. The one-way ANOVA hypothesis

test requires three main assumptions including: (a) samples are independent; (b) each group of samples follows a normal distribution; and (c) homoskedasticity: the standard deviation of samples within each group is equal. We also performed a nonparametric version of one-way ANOVA test: Kruskal–Wallis *H*-test, which requires no explicit assumption on the distribution of samples. Both hypothesis tests gave extremely large statistics and extremely small *p*-values, see Table 1. Next, we investigate other factors that are also associated with human perception of fullness.

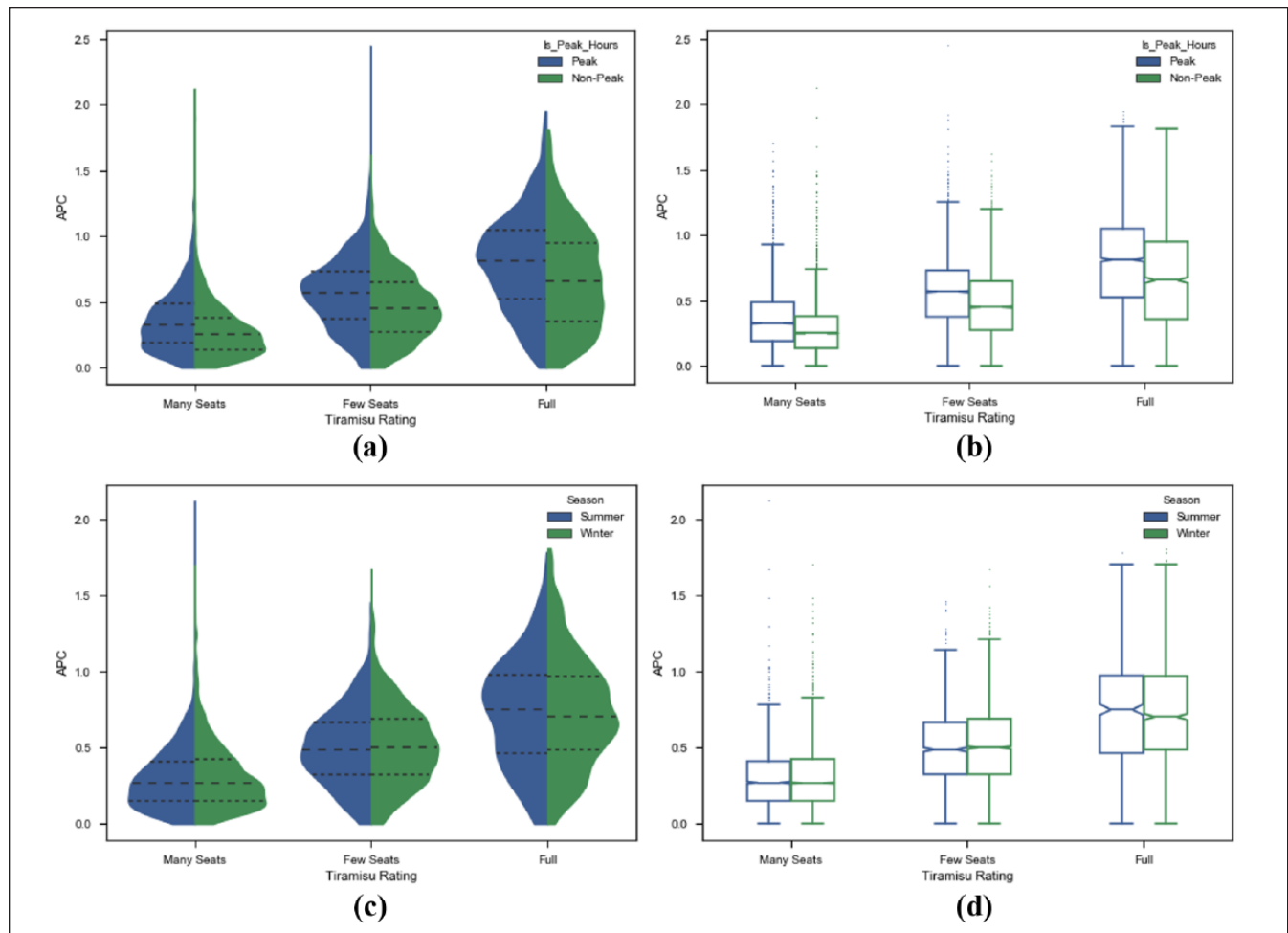
**Figure 3.** Violin plots and box plots of: peak hours versus off-peak hours (a) and (b); winter versus summer (c) and (d).

Table 2. Statistical Tests for the Influence of Time of Day and Seasons

Groups	Mann–Whitney rank test	
	Statistics	p-value
Time of day:		
“Peak-many” versus “off-peak-many”	5.02e+07	7.08e-161*
“Peak-few” versus “off-peak-few”	1.83e+07	3.60e-96*
“Peak-full” versus “off-peak-full”	2.84e+06	4.72e-30*
Seasons:		
“Winter-many” versus “summer-many”	1.05e+07	0.312
“Winter-few” versus “summer-few”	3.27e+06	0.090
“Winter-full” versus “summer-full”	3.78e+05	0.399

Factors Associated with Human Perception of Fullness

For this part, we hoped to find the external factors that significantly affect passengers’ perceptions of bus fullness. Since personal information was not collected in our data set, we focus on the most prominent temporal and spatial impact factors, as well as the bus size.

Temporal analysis. First, we examined the effect of time of day (i.e. “peak hours” versus “off-peak hours”) on the human perception of bus fullness. The “peak hours” are defined as inbound trips (directions from suburbs to downtown) in the morning [7 AM, 10 AM] and outbound trips (directions from downtown to suburbs) in the afternoon/evening [4 PM, 7 PM]. All other trips are treated as “off-peak.”

Figure 3(a and b) shows the distribution of APC for trips of peak hours and off-peak hours, as well as the box plot implying the mean, standard deviation, and outliers of data. We also performed a nonparametric hypothesis test, the Mann–Whitney rank test (Table 2). Passengers’ perceptions of fullness were statistically different in peak hours and off-peak hours, provided with the passenger counts. In peak hours, passengers were more likely to tolerate overcrowding in buses. For all three fullness ratings: “many,” “few,” and “full,” the average APC measurements are higher in peak hours than in off-peak hours.

We also studied seasonal effects, in particular winter and summer, on users’ perceptions of bus fullness. In Pittsburgh, due to the cold climate, people wear bulkier clothes during the winter. We considered December to February as winter, and June to August as summer. Similar to the study of the time-of-day effect, the violin plot and box plots for groups of winter and summer are shown in Figure 3(c–d). Surprisingly, the distributions of APC are quite similar between summer and winter. The Mann–Whitney rank test results in Table 2 also show their group mean is statistically insignificant. Clearly, the data shows that passengers can tolerate the loss of space from thick clothing well.

Spatial Analysis. Next, we examined the spatial variation of fullness perception. It is speculated that geographic regions

with different demographics may have effects on users’ perceptions. We divided the city of Pittsburgh into census tracts, and acquired the median annual household income data for each census tract from 2011 to 2015 using American Community Survey (ACS) 5-Year Estimates. Three income classes are applied.

- Low income: lower than \$40,000
- Middle income: from \$40,000 to \$100,000
- High income: higher than \$100,000

The violin plot and box plot are shown in Figure 4 (a–b). The mean APC measurements for all three rating groups increase along with the income level. These results are interesting, implying that trips in high-income tracts can probably accommodate more passengers for the same fullness rating than trips in low-income tracts. This may be a region-specific result since peak-hour routes serving workers in the finance and legal professions are very busy in Pittsburgh due to constrained parking in the civic center. Hypothesis test results listed in Table 3 support this conclusion. The mean of APC measurements varies substantially among tracts with different income levels, provided with the same fullness ratings.

Bus Seat Capacity. Bus seat capacity may also affect users’ perceptions of fullness. There are two types of buses running in Pittsburgh: small-size buses have 37, 40, or 42 seats, while large-size buses have 56, 57, 60, or 63 seats. The violin plot and box plot are shown in Figure 4(c–d), paired with hypothesis test results in Table 3. The results show that large-size buses have a smaller mean of passenger-over-capacity ratios in each of the rating groups. This effect is statistically significant, and most prominent for those trips rated at “full.” Users with frequent experience of riding small-size buses are more likely to tolerate overcrowding.

Estimating Human Perception of Bus Fullness

We found that human perception of bus fullness is statistically related to time of day (peak hours or off-peak hours), spatial variation (represented by income level) and bus seat

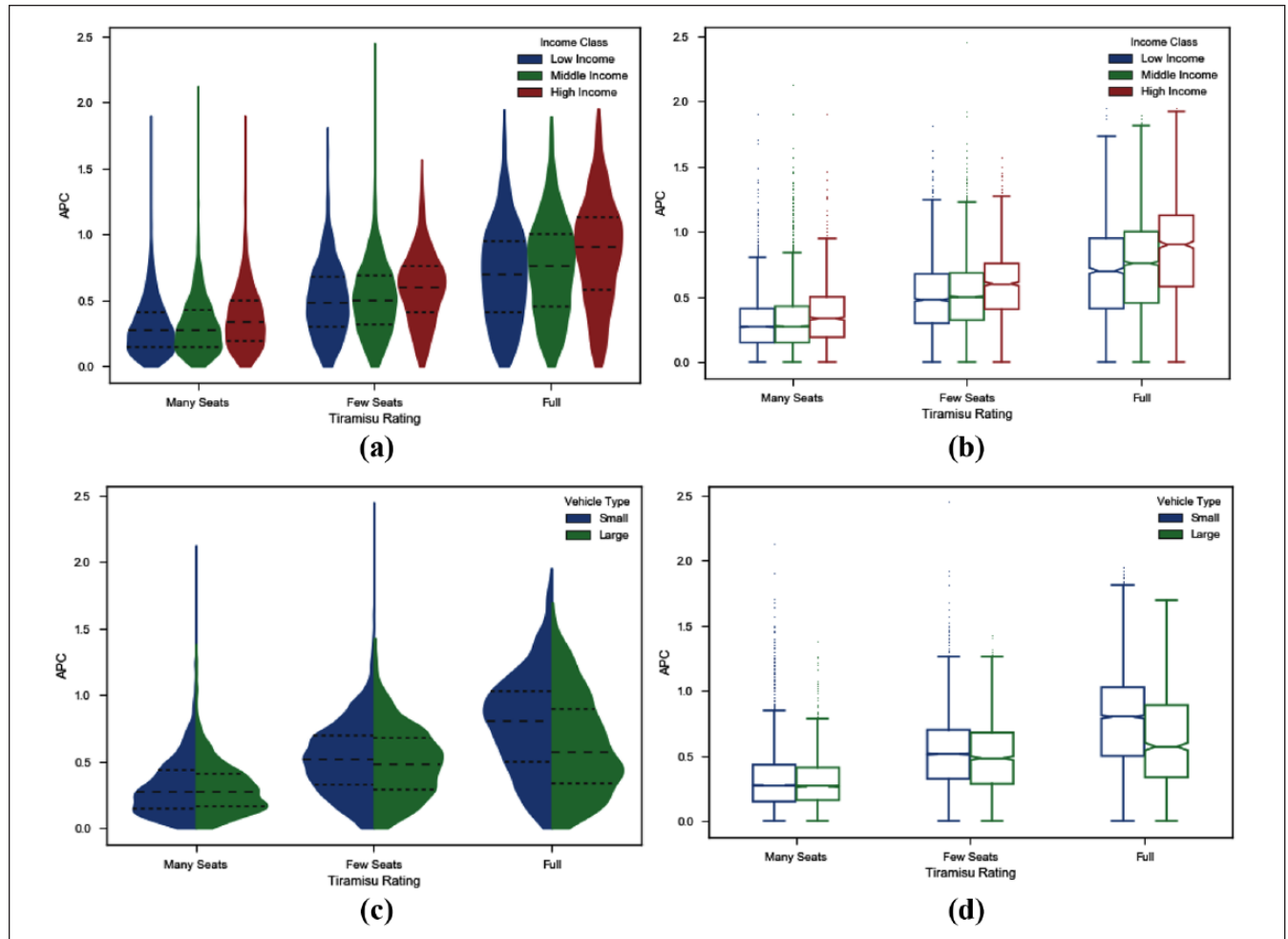


Figure 4. Violin plots and box plots of: different income levels (a) and (b); different vehicle sizes, (c) and (d).

capacity, in addition to the actual passenger-over-capacity ratios measured by APC. This section develops models to estimate human-rated bus fullness using APC data, so that they can be used for transit agencies to assess service quality better without necessarily having crowdsourced ratings data. We defined two classes of models:

- Baseline model: APC-based fullness being the sole predictor
- Improved model: The predictors include APC-based fullness, bus route ID, hour of day, trip direction, median income of the tract, bus seat capacity

Table 3. Statistical Tests for the Influence of Different Income Levels and Vehicle Sizes

Groups	Mann–Whitney rank test <i>p</i> -value
income levels:	
“Low income-many” vs. “middle income-many”	5.14e-03*
“Middle income-many” vs. “high income-many”	2.20e-27*
“Low income-few” vs. “middle income-few”	3.81e-04*
“Middle income-few” vs. “high income-few”	1.07e-31*
“Low income-full” vs. “middle income-full”	2.13e-05*
“Middle income-full” vs. “high income-full”	4.41e-19*
Vehicle sizes:	
“Small-many” vs. “large-many”	0.081
“Small-few” vs. “large-few”	2.96e-04*
“Small-full” vs. “large-full”	4.80e-33*

Table 4. Estimation Accuracy with Different Hyperparameters in a Random Forest Classifier

<i>Min-split</i>	2			3			5			10		
<i>n-trees</i>	B	I	Imp	B	I	Imp	B	I	Imp	B	I	Imp
20	64.0	75.6	18.1%	64.2	75.5	17.7%	64.1	75.2	17.4%	64.0	74.0	15.5%
50	64.0	76.2	19.0%	64.0	76.0	18.7%	64.1	75.4	17.7%	64.0	74.0	15.5%
100	64.0	76.5	19.4%	64.0	76.5	19.4%	64.1	76.1	18.7%	64.1	74.4	16.1%
200	64.1	76.4	19.0%	64.0	76.6	19.6%	64.1	76.1	18.8%	64.0	74.3	16.0%
300	64.0	76.4	19.4%	64.0	76.5	19.4%	64.0	76.1	18.8%	64.1	74.3	15.9%
500	64.0	76.4	19.3%	64.1	76.6	19.5%	64.0	76.3	19.1%	64.1	74.4	16.1%

Note: B = baseline model; I = improved model; Imp = improvement in estimation accuracy.

We randomly selected 50% of the data as the training set for a random forest classifier. The remaining 50% of the data was used as the test data set. We did a grid search of hyperparameters for a random forest classifier: (a) *n-trees*: the number of trees in the forest; and (b) *min-split*: the minimum number of data samples required to split an internal node. The estimation accuracies are shown in Table 4. Figure 5 shows the confusion matrix of baseline and improved model for a random forest classifier with the best hyperparameters found in grid search (*n-trees* = 200, *min-split* = 3).

Generally, the non-linear classification model can accurately estimate the users' fullness ratings 77% of the time. This substantially improves on the baseline model that simply regards the fullness ratings to be equivalent to categorized passenger-over-capacity ratios, where the number is 64%. This also implies that humans' perception of fullness can be influenced by many factors other than the passenger counts.

Bus passengers are heterogeneous, and their perceptions can also be influenced by their personal characteristics. This kind of information is not available and thus cannot be included in our model. The model has an error rate of about 23%. Its accuracy may be further improved by incorporating personal demographic data.

Discussion

Analysis of APC data and crowdsourced fullness rating data revealed that they are highly correlated. On average, users perceive the same fullness that is largely determined by passenger-over-capacity ratios. However, there exist large variance of perception of fullness among users, especially when the bus is crowded. Such variance of perception can be partially attributed to peak hours, bus size, and riders' income level. However, there is no clear seasonal effect on the perception. We summarize additional insights as follows:

- Fullness ratings by users, namely their perception of fullness, can vary quite substantially among users. Therefore, averaged ratings may not reveal user diversity. Generally,

this variation of perception is relatively small when rating a crowded bus during peak hours or an empty bus.

- Riders are more likely to "tolerate" overcrowding during peak hours than off-peak hours. This is probably because riders are used to and prepared for crowded buses during peak hours. For transit agencies, the implication is that when planning and operating bus trips during peak hours, more passengers may be tolerated to ensure the same service quality to users than during off-peak hours.
- Bus trips in high-income areas can probably accommodate more passengers for the same fullness rating than trips in low-income areas. This is probably due to two reasons: self-selection of high-income riders and more space requirement for low-income riders. People with high income are more likely to travel using private cars, so whoever chooses to ride a bus is more likely to tolerate in-vehicle crowding. Low-income riders are likely to be captive to buses. Thus, they may travel with more bags and carry-ons.
- On average, passengers perceive around 90% of seat capacity to be "full" for small-size buses, and 60–70% of capacity to be "full" for large-size buses. Large-size buses have a smaller mean of passenger-over-capacity ratios for each of the three rating groups. This effect is the most prominent for buses perceived as "full." Therefore, there is declining benefit in perceived fullness when extending the number of seats in larger, articulated buses. One possible explanation is that passengers can only get on board at the front door of the bus and they tend not to move to the far back of large-sized buses. Thus, the perceived bus fullness from passengers who have just boarded could be "full" even when the APC measure is only 60–70%.

We built an effective model for estimating passengers' perceptions of fullness using spatial and temporal factors in addition to APC data. Estimating human perceptions of bus fullness solely based on passenger counts is generally inaccurate (64% in our study). If we add trip information (time of day, direction, number of bus seats, route ID) and demographic

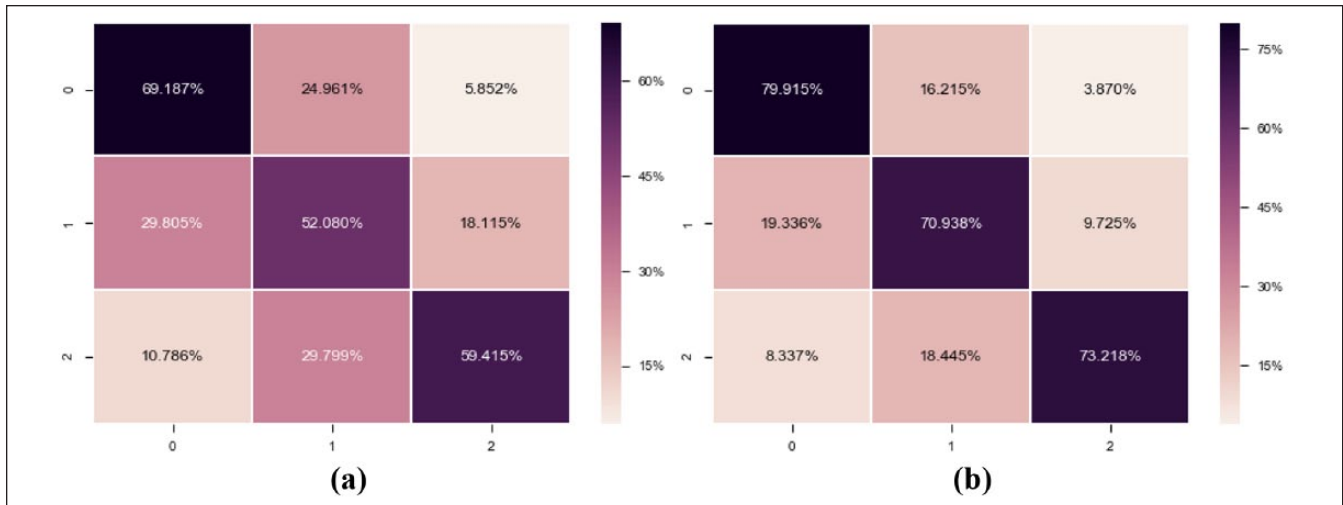


Figure 5. Confusion matrices with heat-maps of baseline and improved models for random forest: (a) baseline model, (b) improved model. Row = prediction label; column = true label; 0 = "many seats"; 1 = "few seats"; 2 = "full".

information (district average annual income) to the predictor, the estimation accuracy can be substantially improved (77% in our study). However, there still exists quite a bit of variability of perception that cannot be explained by the estimation model. This is likely due to individual variability in perception by humans and possibly reckless (false) ratings by some users we were unable to filter out. This noise and error would be extremely difficult to eliminate.

Conclusion and Future Work

This paper examines the linkage between APC data and crowdsourced fullness ratings data from users (acquired from the Tiramisu application) in the Pittsburgh region, to understand better how riders' perceptions of bus fullness are related to spatial, temporal, and demographic factors in addition to passenger counts. We found that human perception of bus fullness, provided with the same passenger counts, can vary substantially by time of day (e.g., peak hours versus off-peak hours), bus vehicle seat capacity (e.g., 40-seats versus 60-seats), and area (differentiated by income level). A model was therefore developed to estimate passengers' fullness ratings using those factors. This model estimates the fullness perception, as a metric for service quality, better than using passenger count as the sole estimator. This model could be used by transit agencies which have APC data to assess service quality better, without necessarily having perception data, and ultimately to enhance system design and operation. The output could also be used to populate fullness rating in real-time feeds for rider information systems. For example, riders who use wheelchairs may opt to alter their trip preemptively if their desired bus is full (7, 10).

Our work has two major limitations. First, we only used transit data from Pittsburgh to build and test the model. Future work should focus on testing this model in other cities and validation using state-preference surveys to assess causality.

Second, our findings suggest that a variety of factors are associated with human perception of bus fullness, but we cannot claim the causality. Future work should apply qualitative methods to investigate the causal relations further. Third, this work could be extended from bus to light rail trains or subways. This study is focused on bus transit due to the limited crowdsourced passenger rating data for light rail transit collected by Tiramisu in Pittsburgh, but this method is applicable if future data for light rail trains becomes available.

Acknowledgments

This research is funded in part by Traffic 21 Institute and Carnegie Mellon University's Mobility 21, a National USDOT University Transportation Center for Mobility which is sponsored by the US Department of Transportation. Parts of this work are also funded under a grant from the National Institute on Disability, Independent Living, and Rehabilitation Research (NIDILRR grant number 90RE5011-01-00). NIDILRR is a Center within the Administration for Community Living, Department of Health and Human Services. The authors also thank Amy Silbermann from the Port Authority of Allegheny County (PAAC) for providing all the APC data for the experiments and analysis.

Author Contributions

The authors confirm contribution to the paper as follows: study conception and design: Xidong Pi, Zhen (Sean) Qian, Yun Huang, Aaron Steinfeld; data collection: Yun Huang, Aaron Steinfeld; data analytics: Xidong Pi, Zhen (Sean) Qian; draft manuscript preparation: Xidong Pi, Zhen (Sean) Qian, Yun Huang, Aaron Steinfeld. All authors reviewed the results and approved the final version of the manuscript.

References

1. dell'Olio, L., A. Ibeas, and P. Cecin. The Quality of Service Desired by Public Transport Users. *Transport Policy*, Vol. 18, No. 1, 2011, pp. 217–227.

2. Eboli, L., and G. Mazzulla. Service Quality Attributes Affecting Customer Satisfaction for Bus Transit. *Journal of Public Transportation*, Vol. 10, No. 3, 2007, p. 2.
3. Lin, T. M., and N. H. Wilson. Dwell Time Relationships for Light Rail Systems. *Transportation Research Record: Journal of the Transportation Research Board*, 1992. 1361: 287–295.
4. Litman, T. Valuing Transit Service Quality Improvements. *Journal of Public Transportation*, Vol. 11, No. 2, 2008, p. 3.
5. Hodges, C. C. *Automatic Passenger Counter Systems: The State of the Practice*. Technical Report No. DOT-1-87-36. Urban Mass Transportation Administration, US Department of Transportation, 1985.
6. Attanucci, J., and D. Vozzolo. Assessment of Operational Effectiveness, Accuracy, and Costs of Automatic Passenger Counters. *Transportation Research Record: Journal of the Transportation Research Board*, 1983. 947: 15–23.
7. Zimmerman, J., A. Tomasic, C. Garrod, D. Yoo, C. Hiruncharoenvate, R. Aziz, N. R. Thiruvengadam, Y. Huang, and A. Steinfeld. Field Trial of Tiramisu: Crowd-Sourcing Bus Arrival Times to Spur Co-Design. *Proc., SIGCHI Conference on Human Factors in Computing Systems*. ACM, 2011, pp. 1677–1686.
8. *Tiramisu: The Real-Time Bus Tracker*. <http://www.tiramisu-transit.com/>. Accessed April 20, 2017.
9. Tomasic, A., J. Zimmerman, C. Garrod, Y. Huang, T. Nip, and A. Steinfeld. The Performance of a Crowdsourced Transportation Information System. *Transportation Research Board 94th Annual Meeting Compendium of Papers*, 2015.
10. Steinfeld, A., J. Zimmerman, A. Tomasic, D. Yoo, and R. Aziz. Mobile Transit Information from Universal Design and Crowdsourcing. *Transportation Research Record: Journal of the Transportation Research Board*, 2011. 2217: 95–102.
11. Cantwell, M., B. Caulfield, and M. O'Mahony. Examining the Factors that Impact Public Transport Commuting Satisfaction. *Journal of Public Transportation*, Vol. 12, No. 2, 2009, p. 1.
12. Milkovits, M. Modeling the Factors Affecting Bus Stop Dwell Time: Use of Automatic Passenger Counting, Automatic Fare Counting, and Automatic Vehicle Location Data. *Transportation Research Record: Journal of the Transportation Research Board*, 2008. 2072: 125–130.
13. Fritz, M. S. Effect of Crowding on Light Rail Passenger Boarding Times. *Transportation Research Record: Journal of the Transportation Research Board*, 1983. 908.
14. Ma, Z., J. Xing, M. Mesbah, and L. Ferreira. Predicting Short-term Bus Passenger Demand Using a Pattern Hybrid Approach. *Transportation Research Part C: Emerging Technologies*, Vol. 39, 2014, pp. 148–163.
15. Tirachini, A., D. A. Hensher, and J. M. Rose. Crowding in Public Transport Systems: Effects on Users, Operation and Implications for the Estimation of Demand. *Transportation Research Part A: Policy and Practice*, Vol. 53, 2013, pp. 36–52.
16. Li, Z., and D. A. Hensher. Crowding in Public Transport: A Review of Objective and Subjective Measures. *Journal of Public Transportation*, Vol. 16, No. 2, 2013, p. 6.
17. Mahudin, N. D. M., T. Cox, and A. Griffiths. Measuring Rail Passenger Crowding: Scale Development and Psychometric Properties. *Transportation Research Part F: Traffic Psychology and Behaviour*, Vol. 15, No. 1, 2012, pp. 38–51.
18. Furth, P. G., B. Hemily, T. H. Muller, and J. G. Strathman. *Using Archived AVL-APC Data to Improve Transit Performance and Management*. Transit Cooperative Research Program (TCRP) Report 113, published by Transportation Research Board of the National Academies, Washington, D.C., 2006.
19. Kittelson & Associates, United States. *Federal Transit Administration, Transit Cooperative Research Program, & Transit Development Corporation. Transit Capacity and Quality of Service Manual* (No. 100). Transportation Research Board of the National Academies, Washington, D.C., 2003.
20. *Mapbox: Build Experiences for Exploring the World*. <https://www.mapbox.com/>. Accessed April 20, 2017.

The Standing Committee on Transit Capacity and Quality of Service (AP015) peer-reviewed this paper (18-05900).

The contents of this paper reflect the views of the authors, who are responsible for the facts and the accuracy of the information presented herein. The U.S. Government and PAAC assume no liability for the contents or use thereof.